

SpaceTEE

Secure and Tamper-proof Computing
in Space using CubeSats

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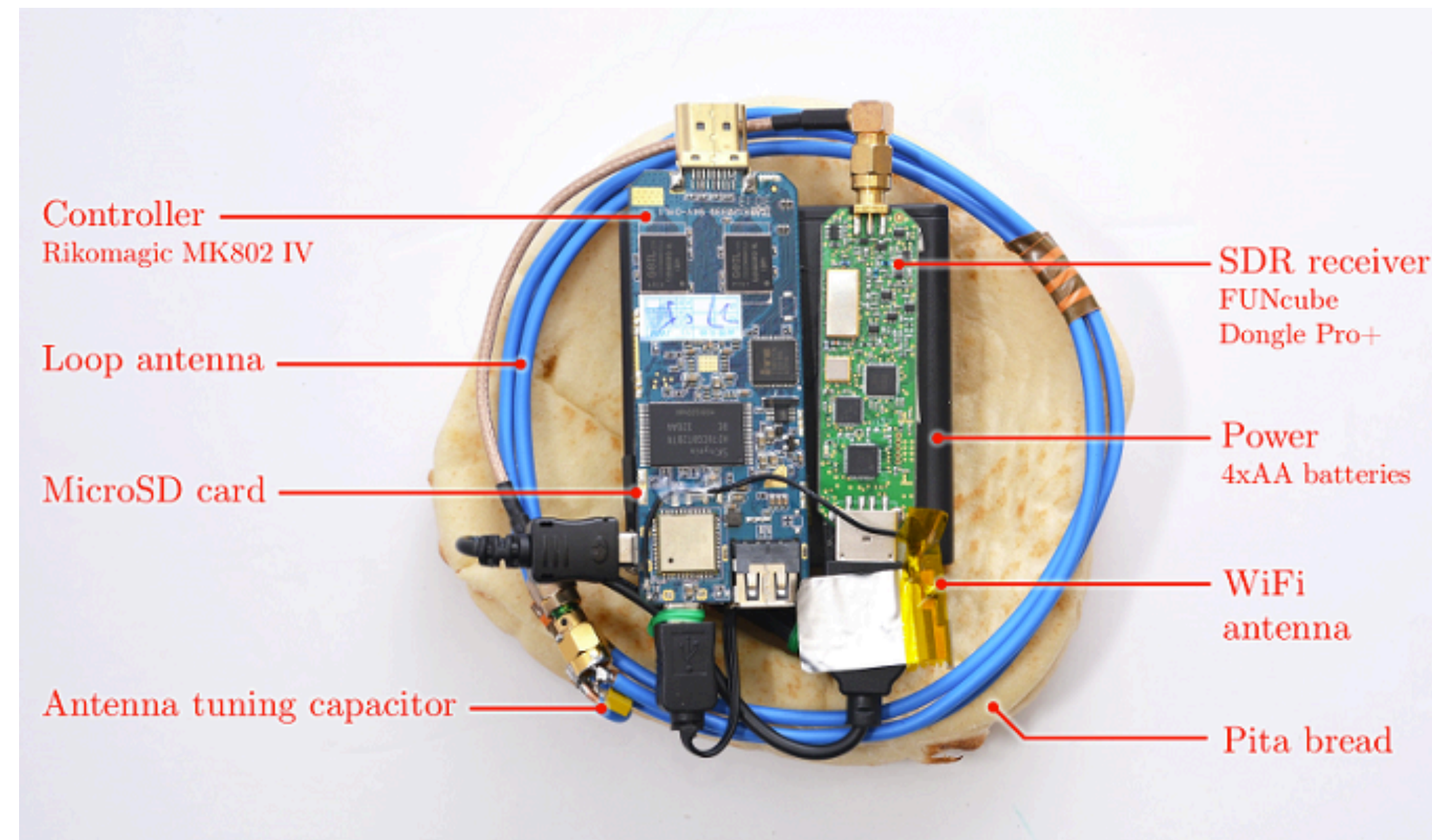
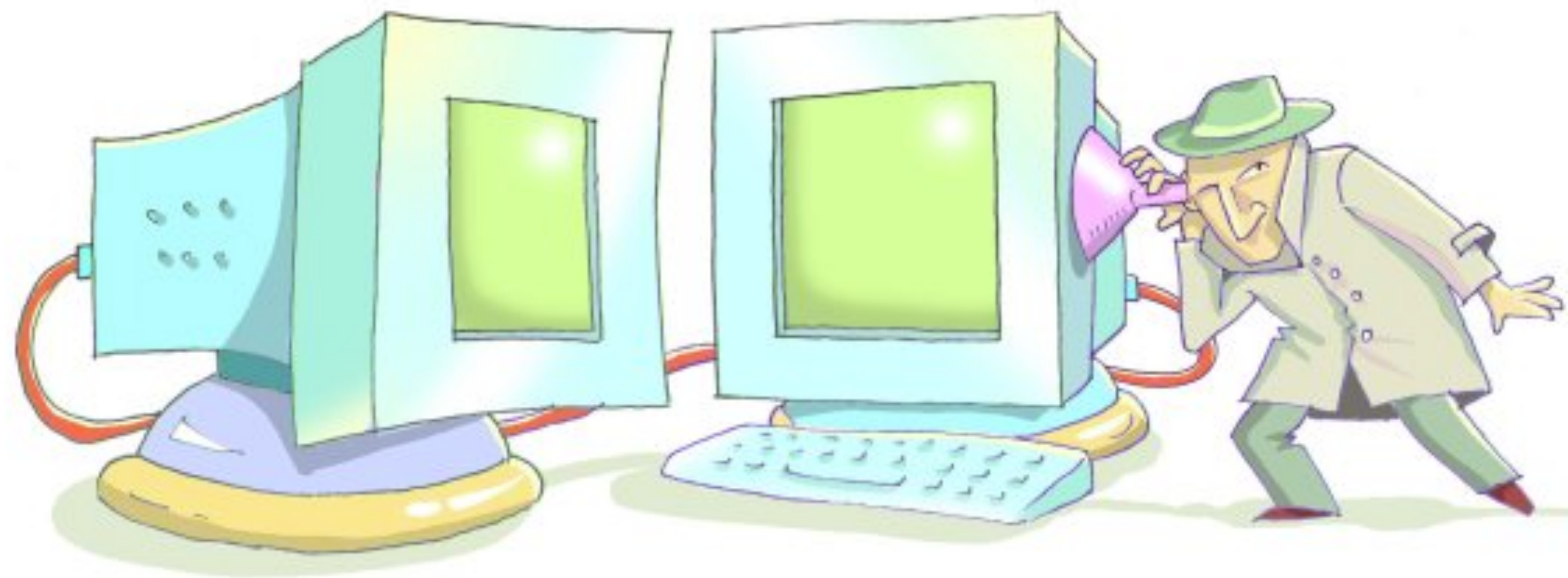


We need TEEs

- For confidentiality
 - Need to make sure inner state is invisible
- For computational integrity
 - Use as a trusted authority
 - Verifiable computation is not yet practical
- Applications: Key management, Authentication, Digital Rights Management, Prevent cheating in online games, Trustworthy grid computing (prevent laziness)

Side-channel attacks

- Physical access (contact) to the device [Genkin et al.'16]
- Key extraction from EM radiation [Genkin et al. '15]
- Acoustic cryptanalysis [Genkin et al. '13]



Hardware Security Modules

- Trusted hardware for crypto operations
- Isolated from the rest of the infrastructure
- Different levels of protection including side-channels and physical access
- Expensive: FIPS 140-2 Level 4 device can cost \$100,000



Whatever you do - someone will probably break it

So what on earth can we do?

Perhaps not on Earth?

- Objects in the outer-space are hard to access physically
- However, they can communicate with one another, and with terrestrial base-stations
- Anti-satellite weapons are mostly able to destroy (DoS), capturing is extremely difficult and expensive, and likely evident
- NORAD: continuous high-res (🏟) monitoring of objects in space using radars and telescopes

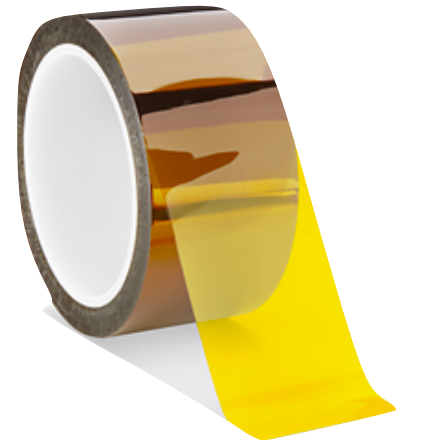
Space is our (🍷) TEE

CubeSats

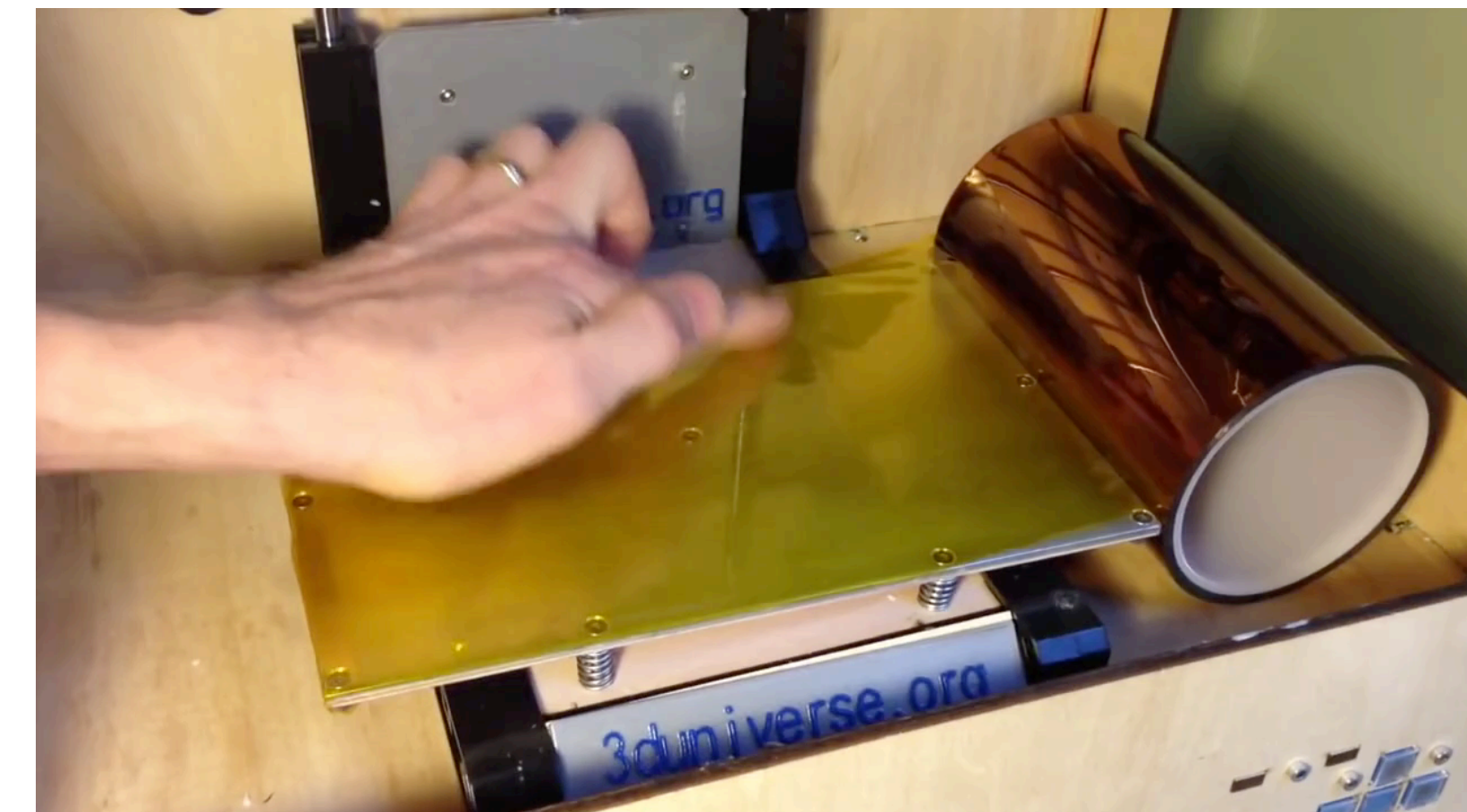


- CubeSat 1U
 - Cheap commercial off-the-shelf nano-satellite $10cm \times 10cm \times 10cm$
 - Weight: 1.33 kg , Size:
 - 699 CubeSats launched (as of June 24th 2017)
 - Our solution estimated cost (including launch) is \$95,000
 - Comparable to a commercial Level 4 HSM

Pre-launch protection



- Make it tamper-evident
 - Cover using Kapton tape and measure pre-launch moment of inertia
 - Compare to observed once in orbit (by tracking the antennas spin via Doppler effect)
 - Tampering with coated hardware without changing it is hard

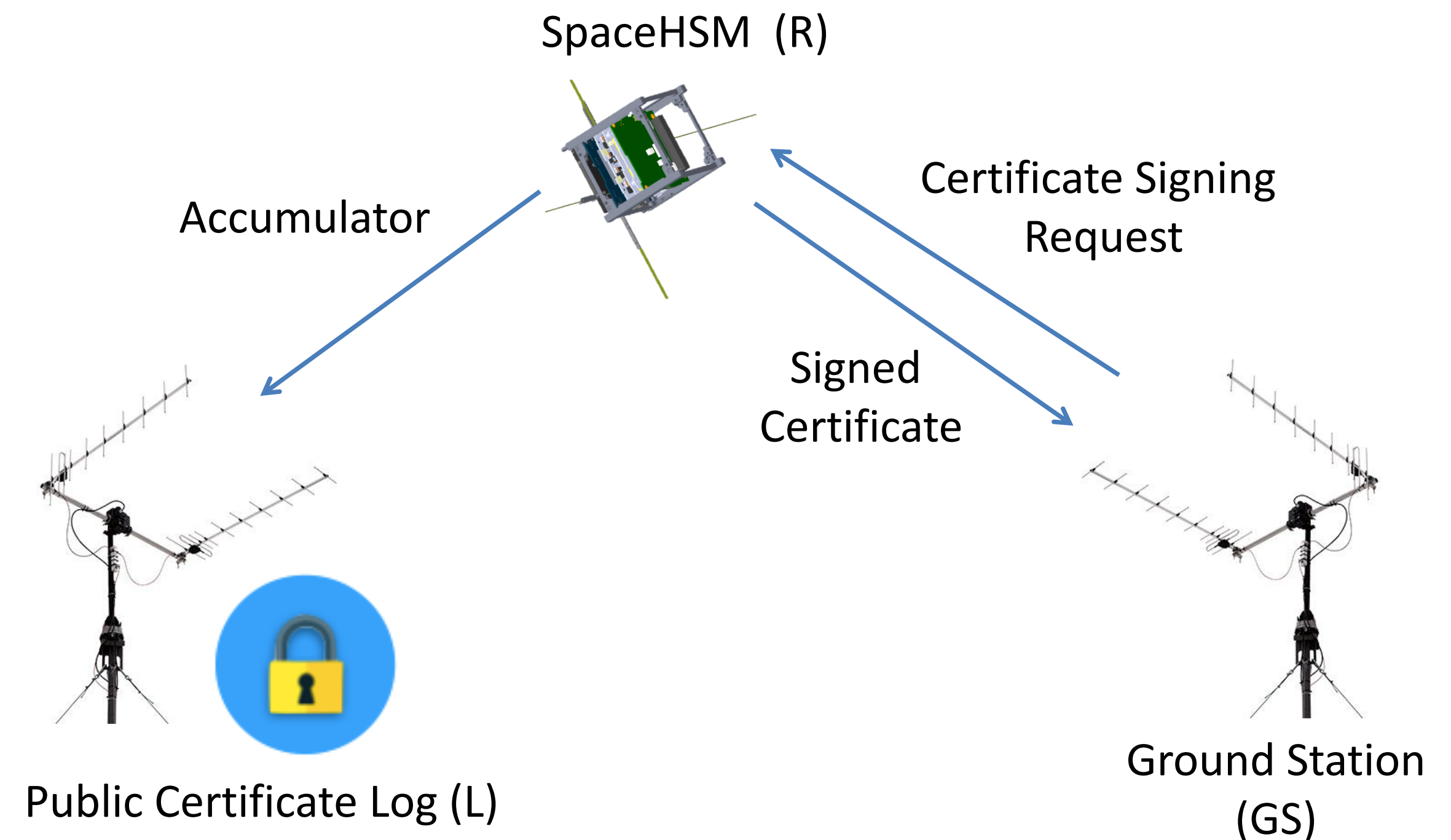


HSM using SpaceTEE: SpaceHSM

- A system that can serve as a Certificate Authority (CA)
- Builds on traditional PKI and Certificate Transparency
- Certificate transparency logs prevent a powerful attacker who gains access to the communication channel with the satellite from obtaining forged certificates

System Overview

1. SpaceHSM: Root-of-Trust (R) and crypto-accumulator
2. Ground-station (GS): delegates cert. sign. requests
3. Public, append-only certificate log (L)
4. Verifier (V): checks certificate validity



Threat Model

- SpaceHSM is tamper-proof. No physical access.
- Adversary:
 - Can physically access to all terrestrial infrastructure
 - Can occasionally access the communication channel with the SpaceHSM, bypassing the security of the ground-station
 - Full read-access to code and data of the SpaceHSM prior to launch
- Spoofing SpaceHSM transmission is impossible

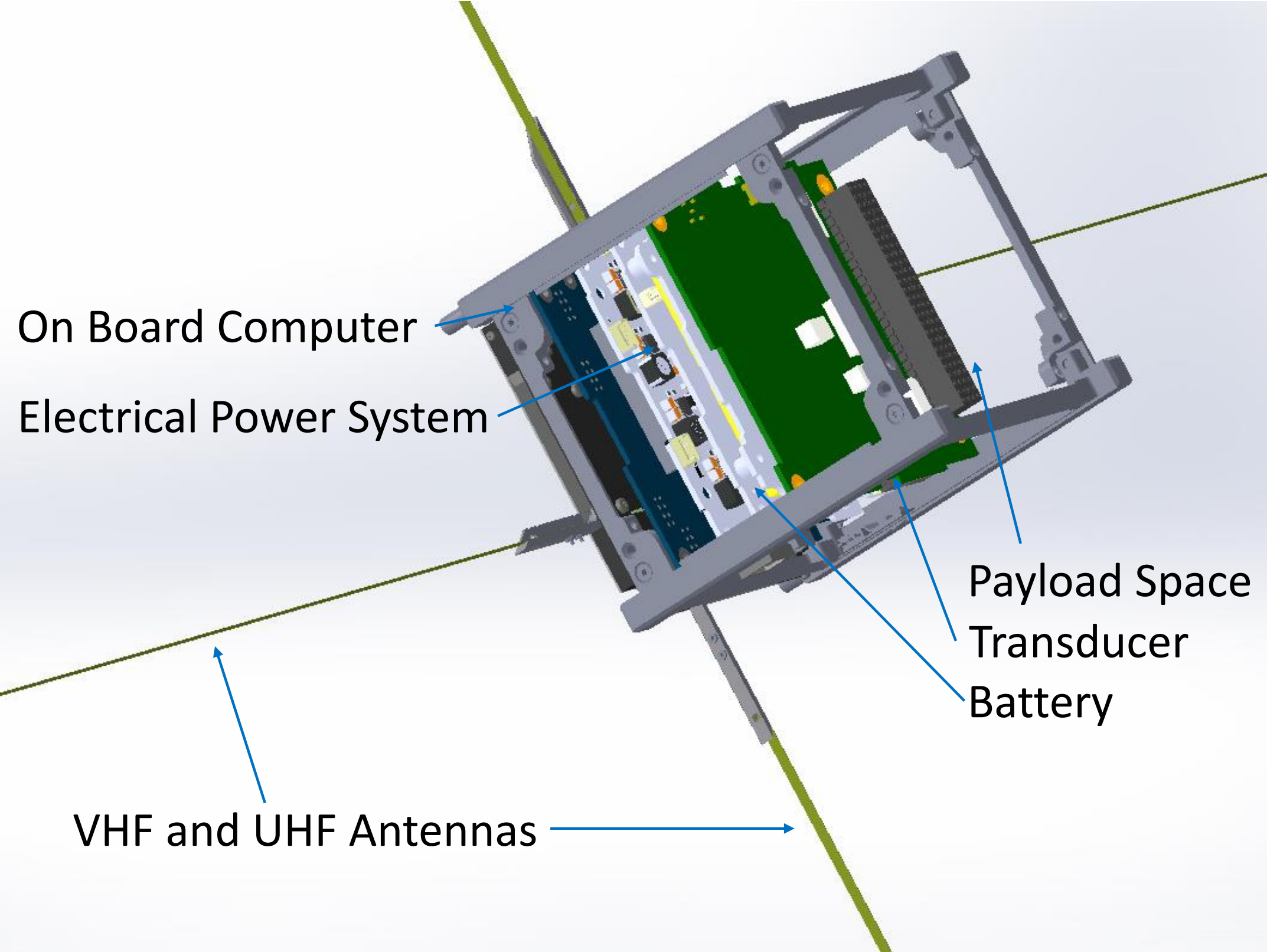
Protocol

- **SpaceHSM Bootstrap:** Generate key-pair and broadcast the public key.
- **Certificate Request:** Requests are transmitted to the SpaceHSM
 - The SpaceHSM transmits back a signed certificate, **updates its accumulator and broadcasts it.**
- **Cert. Log Update:** Every signed certificate is expected to be submitted to the certificate log. The log-server verifies the certificate and appends it.

DoS Attacks

- How about an adversary that attempts to mount a denial-of-service attack?
 - Submit a certificate signing request to SpaceHSM
 - Withhold the signed certificate from the log
- SpaceHSM can broadcast the signed certificate instead of sending it only to the ground-station via the private channel
- Additional measures based on key-rotation are in the paper

CubeSat Design

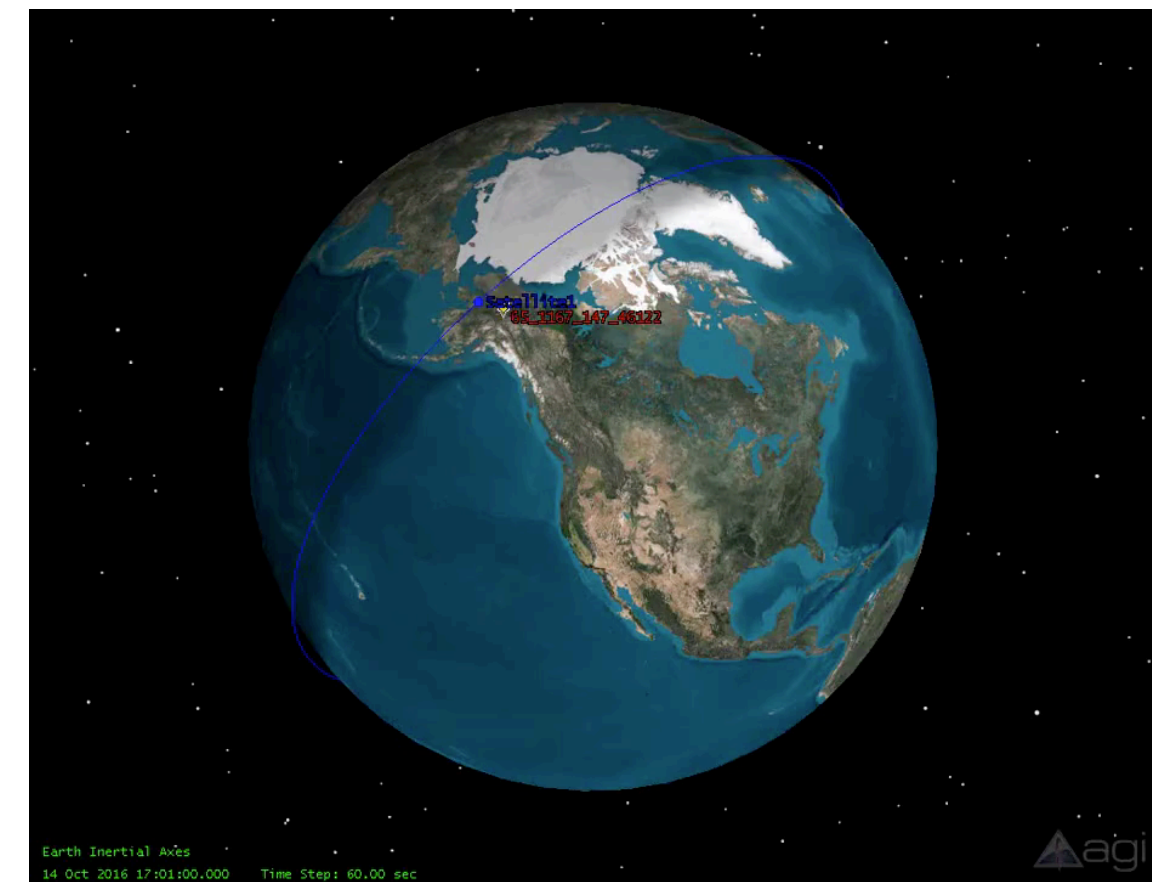


Component	Model Name	Notes
Structure	1U CubeSat	
OBC	Cube Computer [6]	ARM Cortex-M3
Power system	CS 1U Bundle A	10 Whr battery
Solar panels	Azurspace panels [5]	On all 6 sides
Transceiver	ISIS VHF/UHF [17]	Full-duplex
Antennas	ISIS dipole	

Can accommodate 2 independent SpaceHSMs with shared power supply

CubeSat Design

- Use commercial, tested off-the-shelf parts
- Launch onto a Sun Synchronous Orbit (SSO)
- Cube computer:
 - 32-bit ARM Cortex-M3, 4MB Flash memory for code
 - 2 x 1MB SRAM memory for data
 - Connected to 48MHz clock



Power Budget

- 45 min. “daylight”, 45 min. “nighttime”
- Average SpaceHSM power consumption: ~1.7 Watt
- 6 solar panels can generate ~3.8 Watt
- Single power supply can accommodate 2 SpaceHSMs

Link Budget

- Currently expect an uplink of 1200bps
- Given an average certificate size of 2.5KB, we can service one certificate signing request every 20 seconds
- For example, using the Svalbard ground-station, can communicate for 10 min. every 90 min.
- Rates can be improved by 1000x using S-Band transceivers supporting 2Mbps uplink
 - Enables handling 50 cert. req. / sec.

Software

- Requirements:
 - [Symmetric encryption](#): AES-128, AES-256
[PKI](#): RSA-2048, RSA-4096, ECDSA sigs., Curve25519
 - [Small crypto library](#)
- Alternatives:
 - TweetNaCl: 11 KB, 40 RSA-2048 dec./sec. on ARM Cortex M3,
 - mbedTLS: 125 KB (still within RAM budget)
 - SharkSSL: 15 KB (benchmarked on ARM Cortex M3)
 - WolfSSL: small, open-source, supports TLS 1.3, ChaCha20, Curve25519
- [Protect against faults](#) (EDAC hardware, verify signatures)

More applications

- Trusted public parameters generator for crypto schemes
- Trusted party for cryptographic protocols
- Trusted time-stamping
- Trusted mining of cryptocurrencies (transaction verification)

SPACE, THE FINAL FRONTIER

Thank you. Questions please.

